

Java2: Lesson 10 – Lab Project

1. Add **summary** at the top, **documentation** at the bottom and **comments** where necessary.
- Recordings and interviews are an exception, but they should be included in your submission code.
2. **Cite** the source of your information.
3. Refrain from including anything **you don't fully understand**.

1	100 points 5-min challenge: you will get full credit after you reproduce your work during a class or interview. Prompt the user to enter two numbers, then calculate and print out their multiplication. 1. Create a generic class that can work with different data types: short, int, long, double 2. Write a test code to demonstrate that the generic class can handle all the above data types. *For more information, please visit https://www.youtube.com/watch?v=jUcAyZ5OUm0 <i>*Do not edit your recordings. The act of editing is considered a form of cheating.</i>		
	<table> <tr> <td data-bbox="154 709 300 1100">Sample output</td><td data-bbox="300 709 1549 1100"> <pre>num1? 7 num2? 5 7 x 5 =35 ----- num1? 7.0 num2? 5.0 7.0 x 5.0 =35.0 ----- num1? a done!</pre> </td></tr> </table>	Sample output	<pre>num1? 7 num2? 5 7 x 5 =35 ----- num1? 7.0 num2? 5.0 7.0 x 5.0 =35.0 ----- num1? a done!</pre>
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2	100 points Write a code that uses binary search algorithm to find a value in the given array {10,9,8,7,6,5,4,3,2,1,0}. 1. Implement the following generic method for binary search: public static <E extends Comparable<E>> int biSearch(E[] list, E key) 1. Prompt the user to enter a value to search. <i>*Modify the binary search algorithm in the textbook, ensuring that your implementation is different from the one provided. Avoid using any built-in methods, such as 'binarySearch()'.</i>		
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